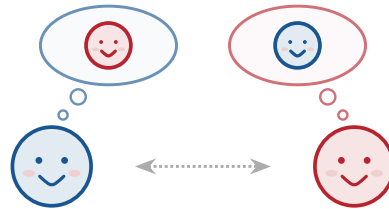


Chapter 7

Rationalizability and Iterated Dominance



Rationality and Beliefs

A rational player chooses a best response to her belief about the other player's strategy.

In a finite two-player game, a pure strategy is a best response to some belief if and only if it is not strictly dominated by any pure or mixed strategy.

Knowing that the other player is rational restricts which strategies a belief can assign positive probability to.

Discussion: What Can Player 1 Choose?

	X	Y	Z
A	3, 3	0, 5	0, 4
B	0, 0	3, 1	1, 2

The players choose simultaneously: player 1 chooses A or B , and player 2 chooses X , Y , or Z . Payoffs are listed in that order.

Watson calls this a “mind-games” example: each player puts herself in the opponent’s position and reasons about what a rational opponent would choose.

Find a belief that makes A optimal and another that makes B optimal. Does this establish that both strategies survive reasoning about player 2’s rationality?

Answer: Rationality with Unrestricted Beliefs

	X	Y	Z
A	3, 3	0, 5	0, 4
B	0, 0	3, 1	1, 2

Against X for sure, A gives $3 > 0$. Against Y for sure, B gives $3 > 0$.

Both rows are best responses to some belief. But we have not yet checked whether those beliefs put weight on choices a rational player 2 would make.

Computing the Best Responses

Let player 1's belief over (X, Y, Z) be $(p, q, 1 - p - q)$, where $p, q \geq 0$ and $p + q \leq 1$. Then

$$u_1(A) = 3p, \quad u_1(B) = 1 - p + 2q.$$

- A is uniquely optimal if $4p > 1 + 2q$.
- B is uniquely optimal if $4p < 1 + 2q$.
- Both are best responses if $4p = 1 + 2q$.

Round 1: Think as Player 2

	X	Y	Z
A	3, 3	0, 5	0, 4
B	0, 0	3, 1	1, 2

Player 2's X is strictly dominated by Y : $5 > 3$ against A , and $1 > 0$ against B . (It is also dominated by Z .)

If player 1 knows player 2 is rational, his belief must put zero probability on X . Thus $p = 0$.

Round 2: Reconsider Player 1

	Y	Z
A	0, 5	0, 4
B	3, 1	1, 2

Once X is excluded, B strictly dominates A : $3 > 0$ and $1 > 0$.

Equivalently, with $p = 0$, playing B gives $1 + 2q > 0$, whereas A gives zero. Player 1's rationality and knowledge of player 2's rationality rule out A .

Discussion: Complete the Argument

	Y	Z
B	3, 1	1, 2

Which column survives now? What must player 2 know to justify ignoring row A ?

Find the surviving payoff vector. Compare it with (A, X) in the original game, which pays $(3, 3)$.

Answer: A Unique, Inefficient Prediction

Against B , player 2 prefers Z to Y because $2 > 1$. Only (B, Z) survives, paying $(1, 2)$.

Player 2 must know that player 1 is rational and knows she is rational. That is what lets her rule out A .

Both players would receive more at (A, X) , which pays $(3, 3)$. Common knowledge of rationality need not produce an efficient outcome.

The Knowledge Behind Each Round

1. Rationality rules out X for player 2.
2. Player 1's rationality and knowledge of player 2's rationality rule out A .
3. Player 2's rationality and knowledge of player 1's reasoning rule out Y .

Common knowledge of the game and rationality supports all these rounds.
This particular example needs only finitely many levels of reasoning.

The General Procedure

Definition. Iterated elimination of strictly dominated strategies (IESDS), or **iterated dominance**, repeatedly removes strictly dominated pure strategies, allowing domination by pure or mixed strategies in the current reduced game.

Start with the full game. Remove dominated strategies for both players, then check dominance again against the remaining opponent strategies.

Stop only when neither player has a strategy left to remove. The rounds describe reasoning before play, not repeated play of the game.

Keeping Track of the Surviving Sets

Write $R_i^0 = S_i$ for player i 's original strategy set and R_i^k for the strategies left after round k . Let $R^k = R_1^k \times R_2^k$.

In the mind-games example, delete column X , then row A , then column Y :

$$R^1 = \{A, B\} \times \{Y, Z\},$$

$$R^2 = \{B\} \times \{Y, Z\},$$

$$R^3 = \{B\} \times \{Z\} = R.$$

We remove choices from consideration. The payoffs of the surviving cells do not change.

Rationalizable Strategies

Definition. In a finite two-player game, a pure strategy is **rationalizable** if it survives iterated removal of strictly dominated strategies, including dominance by mixtures.

Equivalently, repeatedly delete strategies that are never best responses to beliefs over the other player's remaining strategies.

A best response to an unrestricted belief need not be rationalizable. In the mind-games example, A is optimal against X . But X is strictly dominated for player 2; after excluding X , B strictly dominates A .

What Supports a Surviving Strategy?

At the end, each surviving strategy is a best response to some belief supported on the opponent's surviving strategies.

Each opponent strategy in that belief can itself be justified by a belief over our surviving strategies. The justification continues in both directions.

These beliefs need not be correct about actual play, and different strategies may need different supporting beliefs. Rationalizability does not assume that everyone correctly predicts everyone else.

Rationalizable Strategy Profiles

Definition. A strategy profile is **rationalizable** if every player's strategy in that profile is rationalizable.

If R_1 and R_2 are the players' sets of rationalizable strategies, the set of rationalizable profiles is

$$R = R_1 \times R_2.$$

Each strategy must be optimal under some belief about the opponent's surviving strategies. It need not be optimal against the opponent's actual strategy in the profile.

In the mind-games example, $R_1 = \{B\}$ and $R_2 = \{Z\}$, so (B, Z) is the only rationalizable profile.

Discussion: A Mixture Starts the Process

	X	Y	Z
U	5, 1	0, 4	1, 0
M	3, 1	0, 0	3, 5
D	3, 3	4, 4	2, 5

Watson's Figure 7.3 has no strict dominance by a pure strategy initially. For player 2, compare X with a half-and-half mixture of Y and Z .

Calculate the mixture's payoff against each row. Which column can we remove?

Answer: Check Every Opponent Strategy

Player 2's payoffs are

Player 1's row	U	M	D
X	1	1	3
$\frac{1}{2}Y + \frac{1}{2}Z$	2	2.5	4.5

Every comparison is strict, so delete X .

The mixture is a way to demonstrate that X is never optimal. We are not predicting that player 2 actually uses this mixture.

Discussion: Finish the Deletions

	Y	Z
U	0, 4	1, 0
M	0, 0	3, 5
D	4, 4	2, 5

Starting from this reduced game, record each further deletion and the strategy that dominates it.

State the final sets R_1 and R_2 , and the resulting payoff vector.

Answer: Four Rounds in Total

1. Remove X using the mixture of Y and Z .
2. Remove U : D gives $4 > 0$ against Y and $2 > 1$ against Z .
3. Remove Y : against M, D , player 2 gets 5, 5 from Z , exceeding 0, 4 from Y .
4. Remove D : against Z , player 1 gets 3 from M and 2 from D .

Thus $R_1 = \{M\}$ and $R_2 = \{Z\}$. The payoff vector is $(3, 5)$.

When Can We Stop?

In a finite game, every nonterminal round removes at least one strategy, so the deletion process eventually stops.

A unique survivor for each player gives a unique rationalizable profile. Several survivors require a further check: none may be dominated by a mixture either.

To establish that a set survives, find a supporting belief for each strategy. A best response to a belief over the surviving opponent strategies cannot be strictly dominated in that reduced game.

Discussion: Strict Versus Weak Dominance

	L	R
U	1, 1	1, 0
D	1, 0	0, 1

Does U strictly dominate D ? Can a rational player choose D ?

Find a supporting belief for each row and each column. What survives iterated strict dominance?

Answer: Do Not Delete a Best Response

U ties D against L and beats it against R . This is weak dominance, not strict dominance.

D is a best response to L for sure; U is a best response to R for sure. For player 2, L is optimal against U , and R against D .

All four strategies survive. Deleting D would impose a stronger requirement than rationality.

Discussion: The Tennis Game

	F	C	B
F	0, 5	2, 3	2, 3
C	2, 3	0, 5	3, 2
B	5, 0	3, 2	2, 3

In Watson's guided exercise, the server chooses forehand (F), center (C), or backhand (B); the receiver simultaneously chooses which side to favor.

For the server, try probability q on B and $1 - q$ on C . For which values of q does this mixture strictly dominate F ?

Answer: Find the Mixing Interval

Against receiver choices F, C, B , respectively, the mixture gives

$$2 + 3q, \quad 3q, \quad 3 - q.$$

The server's row F gives $0, 2, 2$. Strict dominance requires

$$2 + 3q > 0, \quad 3q > 2, \quad 3 - q > 2.$$

Thus any $\frac{2}{3} < q < 1$ works. For example, $q = \frac{3}{4}$ gives payoffs $4.25, 2.25, 2.25$, strictly above $0, 2, 2$. Delete row F .

Discussion: What Survives in Tennis?

	F	C	B
C	2, 3	0, 5	3, 2
B	5, 0	3, 2	2, 3

Find the receiver's next deletion. Then determine whether any further strategies can be removed.

For each strategy you retain, give a belief over the other player's remaining strategies that makes it a best response.

Answer: Several Strategies Survive

	C	B
C	0, 5	3, 2
B	3, 2	2, 3

Receiver F is dominated by C : $5 > 3$ and $2 > 0$.

In the remaining game, the server prefers B against C , and C against B .
The receiver prefers C against C , and B against B .

Every remaining strategy has a supporting belief. Therefore $R = \{C, B\} \times \{C, B\}$.

Strategic Uncertainty

	Stag	Hare
Stag	5, 5	0, 4
Hare	4, 0	4, 4

In Watson's Stag Hunt (Figure 7.4), hunting Hare guarantees 4. Hunting Stag yields 5 if the other player joins, and zero otherwise.

Both choices are rationalizable for each player. Common knowledge of rationality does not tell us which choice the other player will make.

Discussion: How Much Confidence Is Enough?

	Stag	Hare
Stag	5, 5	0, 4
Hare	4, 0	4, 4

Let p be the probability that the other player hunts Stag. Derive the best-response threshold, including the tie.

Can (Stag, Hare) be a rationalizable profile even though the hunters fail to coordinate? Explain using beliefs.

Answer: Rational Beliefs Can Be Wrong

Stag yields $5p$ and Hare yields 4. Stag is uniquely optimal for $p > 4/5$; Hare for $p < 4/5$. Both are optimal at $p = 4/5$.

At (Stag, Hare), player 1 could expect Stag for sure, while player 2 expects Hare for sure. Each is choosing a best response to a belief supported on rationalizable strategies.

Both beliefs are wrong about actual play. Rationalizability permits this: all four pure profiles survive, including the two mismatches.

Two Different Strategic Tensions

- **Individual incentives and efficiency.** In the mind-games example, only (B, Z) survives, paying $(1, 2)$. Both players prefer (A, X) , which pays $(3, 3)$.
- **Strategic uncertainty.** In Stag Hunt, both players would like the high-payoff coordinated outcome, but uncertainty about the other can make Hare optimal.

More reasoning alone does not remove every coordination problem. If every remaining strategy already has a rational justification, another deletion round changes nothing.

Discussion: Separated at Taipei Main Station

Two friends lose contact and must each choose one meeting point: the central meeting area or a previously used exit. Each gets 1 if they meet and 0 otherwise.

Write the payoff matrix. Which strategies are rationalizable?

Would an agreement made before entering the station help? Would a location that seems obvious to only one friend necessarily help?

Answer: Coordinating Expectations

	Central area	Exit
Central area	1, 1	0, 0
Exit	0, 0	1, 1

Both choices survive for both friends. Each is optimal if the other friend is expected there.

A shared agreement can align their expectations. A **focal point** is a prominent choice that helps people coordinate their beliefs; private salience alone does not ensure agreement.

Suppose the central area has a large clock that both friends recognize as a meeting landmark. Each may head for the clock, expecting the other to do the same, without having agreed in advance.

Communication, Norms, and Institutions

Communication can align beliefs without changing the payoff matrix, as when the friends agree on a meeting point.

A shared convention (e.g., walking on the right side of a hallway) can serve a similar role. It works through what each person expects the other to do.

Institutions (e.g., courts enforcing contracts) can also change incentives through enforcement or penalties. When that happens, we must change the game itself before applying the same reasoning.

Application: Two Bubble Tea Carts

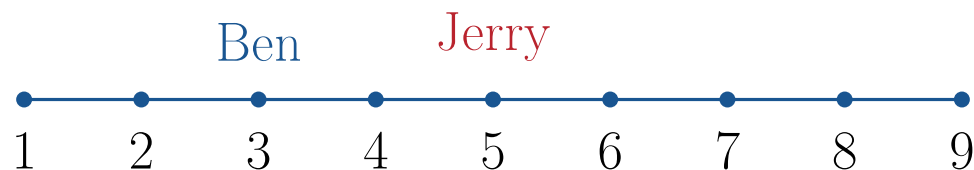
Ben and Jerry simultaneously choose locations on a path with nine equally spaced sections, numbered 1 through 9.

Each section has 100 customers. Everyone buys from the closer cart; an equidistant section splits its customers equally. Carts at the same location split all 900 customers equally.

Prices and margins are identical and fixed, so each cart maximizes its customer count. Both have strategy set $\{1, 2, \dots, 9\}$.

Discussion: Count the Customers

Suppose Ben chooses section 3 and Jerry chooses section 5.



How many customers does each cart serve? What happens if both choose section 5?

When a location is deleted as a strategy, do its customers disappear?

Answer: All Customers Stay in Place

With Ben at 3 and Jerry at 5, Ben serves sections 1, 2, 3 and half of section 4: 350 customers.

Jerry serves the other half of section 4 and sections 5 through 9: 550 customers. At $(5, 5)$, each cart serves 450.

Deleting a location removes a possible cart choice. All 900 customers stay where they are throughout the deletion process.

Why Section 1 Is Strictly Dominated

Compare Ben's customer counts at sections 1 and 2 for every possible location j of Jerry:

Jerry's location	$j = 1$	$j = 2$	$j \geq 3$
Ben at 1	450	100	$50j$
Ben at 2	800	450	$50(j + 1)$

Moving to 2 is strictly better in every case. By symmetry, 8 dominates 9, for either cart.

Round 1 leaves locations $\{2, 3, \dots, 8\}$.

Discussion: Repeat the Location Argument

Now Jerry can choose only $2, 3, \dots, 8$. Compare Ben's section 2 with section 3.

Calculate the customer counts when Jerry is at 2, at 3, and at $j \geq 4$.

Which two locations can both carts remove in round 2? Continue to the final surviving location, keeping all customers in place.

Answer: Move Inward Again

Jerry's location	$j = 2$	$j = 3$	$j \geq 4$
Ben at 2	450	200	$50(j + 1)$
Ben at 3	700	450	$50(j + 2)$

Section 3 strictly dominates 2 in this reduced game. Symmetrically, section 7 dominates 8.

Round 2 leaves $\{3, 4, 5, 6, 7\}$. The same inward comparison removes 3, 7 in round 3 and 4, 6 in round 4.

The Surviving Locations

Round	Locations remaining for each cart
0	1, 2, 3, 4, 5, 6, 7, 8, 9
1	2, 3, 4, 5, 6, 7, 8
2	3, 4, 5, 6, 7
3	4, 5, 6
4	5

Both carts choose section 5 and serve 450 customers each. This is the unique rationalizable profile.

The result uses fixed prices, identical margins, and nearest-cart demand. Different assumptions define a different game.

What the Prediction Assumes

The deductions use common knowledge of both the game and rationality. Knowing our own payoff is not enough to infer the other player's choices.

Uncertainty about a rival's objectives or payoffs can undermine a deletion argument. A mistaken prediction need not mean someone failed to maximize their own payoff.

The equivalence between rationalizability and iterated strict dominance applies to finite two-player games. Infinite strategy sets may require infinitely many rounds; games with more players require care about correlation in beliefs.

Discussion: Check the Logic

Decide whether each statement is true or false and explain.

1. A strategy that survives the first round is rationalizable.
2. If no pure strategy dominates another, deletion must stop.
3. A rationalizable strategy must be optimal against the opponent's actual choice.
4. A unique rationalizable profile must be efficient.

Answer: Four Common Mistakes

1. False. In the mind-games example, row A survives round 1. Removing column X makes B strictly dominate A in round 2.
2. False. A mixture can dominate, as in Figure 7.3 and the tennis game.
3. False. Rationalizability permits incorrect beliefs, as in the mismatched Stag Hunt outcome.
4. False. The mind-games outcome (B, Z) pays $(1, 2)$; both players prefer (A, X) , which pays $(3, 3)$.

Takeaways

Rationalizability requires reasoning about others' rationality.

- Repeatedly remove strictly dominated strategies, allowing domination by mixtures.
- Each surviving strategy has a supporting belief over the opponent's surviving strategies.
- Common knowledge of rationality can yield a unique prediction without guaranteeing efficiency or coordination.
- Several surviving strategies leave strategic uncertainty; supporting beliefs need not be correct.